Project: Worked examples
Title: Psychrometrics
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Calculates the state of moist air from dry-bulb and one other property at a stated pressure or altitude. It then applies one air-handling process with its heat and moisture balance: sensible heating or cooling, dehumidification, humidification or mixing.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Saturation and humidity | $p_{ws}(T)$ by Hyland–Wexler over water at or above 0 °C and over ice below; $W = 0.621945\ p_w/(p - p_w)$; $\varphi = p_w/p_{ws}$ |
| Enthalpy and volume per kilogram of dry air | $h = 1.006\ T + W(2501 + 1.86\ T)$ kJ/kg; $v = 0.287042\ (T + 273.15)(1 + 1.607858\ W)/p$ |
| Dew point and wet-bulb temperatures, by bisection | $p_{ws}(T_{dp}) = p_w$; the wet-bulb from the adiabatic-saturation relation |
| Sensible heating or cooling at constant moisture content | $\Phi = \dot{m}\ (h_2 - h_1)$ |
| Cooling with dehumidification, less the heat leaving in the condensate | $Q_{total} = \dot{m}(h_1 - h_2) - \dot{m}_w\ h_w$, with $\dot{m}_w = \dot{m}(W_1 - W_2)$ |
| Sensible and latent split of the coil load, along $T_1 \rightarrow T_2$ at $W_1$ then $W_1 \rightarrow W_2$ at $T_2$ | $Q_{sensible} = \dot{m}(h_1 - h(T_2, W_1))$; $Q_{latent} = \dot{m}(h(T_2, W_1) - h_2)$; $SHR = Q_{sensible}/(Q_{sensible} + Q_{latent})$; with $RH_2$ blank and $T_2$ above the entering dew point the coil is dry, $W_2 = W_1$ |
| Steam and adiabatic humidification | $h_2 = h_1 + (W_2 - W_1)\ h_{steam}$ for steam; $h_2 = h_1$ for adiabatic (spray) humidification |
| Mixing two airstreams, weighted by dry-air mass flow | $W_{mix} = \Sigma\dot{m}\ W/\Sigma\dot{m}$; $h_{mix} = \Sigma\dot{m}\ h/\Sigma\dot{m}$ |

Basis: Psychrometrics (ideal-gas relations, processes)[1](Ch. 1); Saturation pressure of water and ice[2]; Properties of humid air (principles)[3](§1).

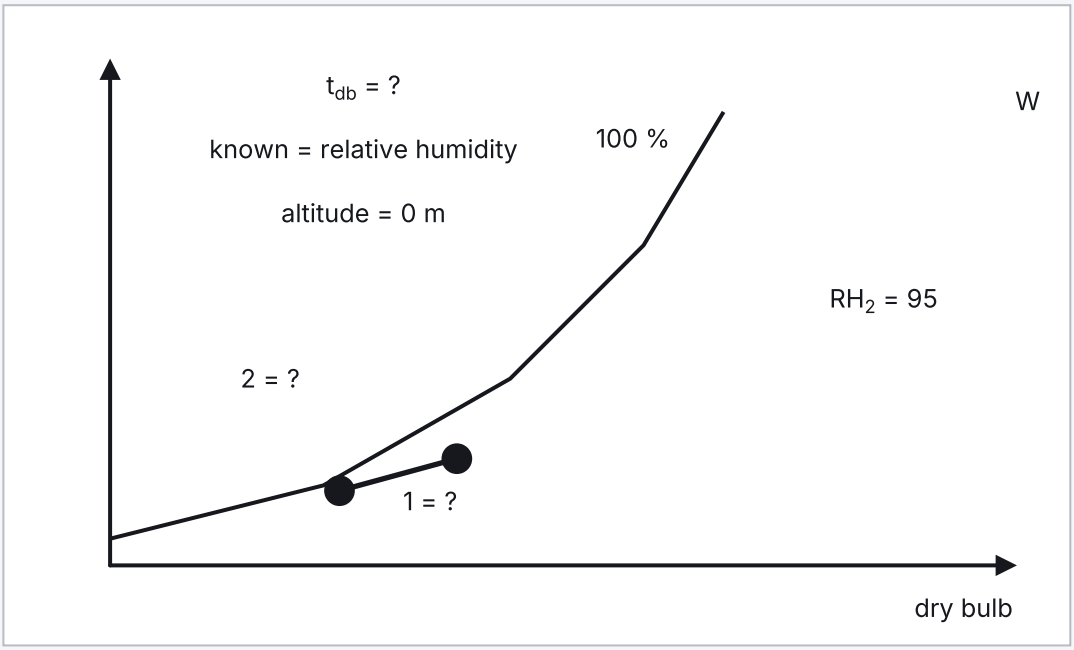



Figure 1 — Cooling with dehumidification: the state cools to $t_2$ at $RH_2$, losing moisture. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| known | Second known property | `relative humidity` | | assumption: RH given. | any |
| process | Process | `cooling with dehumidification` | | assumption: Chilled-water coil. | any |
| altitude | Altitude (used when no pressure is entered) | `0` | m | assumption: Sea-level site. | -500 … 5000 m |
| $T_{db}$ | Dry-bulb temperature | 28 | °C | spec: CIBSE Guide A summer design condition for the site, assumed | -40 … 60 °C |
| RH | Relative humidity | 60 | | spec: Design condition | 5 … 100 |
| $m_{da}$ | Dry-air mass flow | 2.4 | kg/s | spec: AHU schedule AHU-01: 2.0 m³/s ≈ 2.4 kg/s dry air | 0 … 1000 kg/s |
| $h_{steam}$ | Steam enthalpy (steam humidification) | 2676 | | default | 2500 … 3000 |
| $T_2$ | Leaving dry-bulb (sensible, cooling) | 12 | °C | spec: AHU schedule AHU-01 off-coil 12 °C | -40 … 80 °C |
| $RH_2$ | Leaving relative humidity (cooling, evaporative) | 95 | | assumption: Off-coil 95 % RH (typical contact factor). | 5 … 100 |

## 3. Assumptions

A1. Moist air is an ideal mixture of ideal gases. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A2. The enhancement factor is taken as 1. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A3. Condensate leaves at the coil leaving dry-bulb. (Process conventions.)
A4. Steam adds the enthalpy $h_{steam}$ at negligible mass of dry air. (Process conventions.)
A5. Adiabatic humidification follows a line of constant enthalpy, which is close to constant wet-bulb. (Evaporative process idealisation.)
A6. known (Second known property) is taken as relative humidity. (RH given.)
A7. process (Process) is taken as cooling with dehumidification. (Chilled-water coil.)
A8. altitude (Altitude (used when no pressure is entered)) is taken as 0 m. (Sea-level site.)
A9. $RH_2$ (Leaving relative humidity (cooling, evaporative)) is taken as 95. (Off-coil 95 % RH (typical contact factor).)

## 4. Working

[2] $p_{atm}$ = 101.325 (1 - 2.25577e-5 z)^5.2559 [1](Ch. 1 eq. 3)
= z = 0 m
= 101.32 kPa
Barometric pressure (standard atmosphere)
[6] $p_{ws,1}$ = $p_{ws}$(t) Hyland–Wexler (entering)
= t = 28 °C
= 3.7822 kPa
Saturation pressure, state 1
[7] $p_{w,1}$ = p W/(0.621945 + W)
= W = 0.014249
= 2.2693 kPa
Vapour pressure, state 1
[8] $T_{1,out}$ = dry-bulb (entering)
=
= 28 degC
Dry-bulb, state 1
[9] $W_1$ = 0.621945 $p_w$/(p - $p_w$)
= 2.2693 / (101.32 - 2.2693)
= 14.249
Humidity ratio, state 1 (g/kg dry air)
[10] $RH_1$ = $p_w$ / $p_{ws}$
= 2.2693 / 3.7822
= 60
Relative humidity, state 1 (%)
[11] $\varphi_1$ = $p_w$ / $p_{ws}$
= 2.2693 / 3.7822
= 60 %
Relative humidity, state 1, as a percentage quantity (the same value as $RH_1$; a plain-number output cannot be linked to a percentage input)
[12] $T_{dp,1}$ = $p_{ws}$($t_{dp}$) = $p_w$, bisection
= $p_w$ = 2.2693 kPa
= 19.514 degC
Dew point, state 1
[13] $T_{wb,1}$ = adiabatic-saturation relation, bisection
= t = 28, W = 0.014249
= 22.074 degC
Wet-bulb, state 1
[14] $h_1$ = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 28 + 0.014249 × (2501 + 1.86 × 28)
= 64.546

Enthalpy, state 1 (kJ/kg dry air)

[15] $v_1$ = 0.287042 (t + 273.15)(1 + 1.607858 W)/p

= p = 101.32 kPa

= 0.87267 m³/kg

Specific volume, state 1 (per kg dry air)

[16] $\rho_1$ = (1 + W)/v

=

= 1.1622 kg/m³

Moist-air density, state 1

[21] $p_{ws,2}$ = $p_{ws}$(t) Hyland-Wexler (leaving)

= t = 12 °C

= 1.4026 kPa

Saturation pressure, state 2

[22] $p_{w,2}$ = p W/(0.621945 + W)

= W = 0.0082878

= 1.3325 kPa

Vapour pressure, state 2

[23] $T_{2,out}$ = dry-bulb (leaving)

=

= 12 degC

Dry-bulb, state 2

[24] $W_2$ = 0.621945 $p_w$/(p - $p_w$)

= 1.3325 / (101.32 - 1.3325)

= 8.2878

Humidity ratio, state 2 (g/kg dry air)

[25] $RH_2$ = $p_w$ / $p_{ws}$

= 1.3325 / 1.4026

= 95

Relative humidity, state 2 (%)

[26] $\varphi_2$ = $p_w$ / $p_{ws}$

= 1.3325 / 1.4026

= 95 %

Relative humidity, state 2, as a percentage quantity (the same value as $RH_2$; a plain-number output cannot be linked to a percentage input)

[27] $T_{dp,2}$ = $p_{ws}$($t_{dp}$) = $p_w$, bisection

= $p_w$ = 1.3325 kPa

= 11.225 degC

Dew point, state 2

[28] $T_{wb,2}$ = adiabatic-saturation relation, bisection

= t = 12, W = 0.0082878

= 11.553 degC

Wet-bulb, state 2

[29] $h_2$ = 1.006 t + W (2501 + 1.86 t)

= 1.006 × 12 + 0.0082878 × (2501 + 1.86 × 12)

= 32.985

Enthalpy, state 2 (kJ/kg dry air)

[30] $v_2$ = 0.287042 (t + 273.15)(1 + 1.607858 W)/p

= p = 101.32 kPa

= 0.81856 m³/kg

Specific volume, state 2 (per kg dry air)

[31] $\rho_2$ = (1 + W)/v

=

= 1.2318 kg/m³

Moist-air density, state 2

[32] $m_w = \dot{m} (W_1 - W_2)$
$= 2.4 \times (14.249 - 8.2878)$ g/kg
$= 0.014306$ kg/s
Condensate
[33] $h_w = 4.186\ t_2$ (condensate enthalpy)
$= 4.186 \times 12$
$= 50.232$
Condensate enthalpy, kJ/kg
[34] $Q_{total} = \dot{m} (h_1 - h_2) - \dot{m}_w\ h_w$
$= 2.4 \times (64.546 - 32.985) - 0.014306 \times 50.232$
$= 75.027$ kW
Total cooling load
[35] $Q_{sensible} = \dot{m} (h_1 - h(t_2, W_1))$
$= 2.4 \times (64.546 - 48.026)$
$= 39.648$ kW
Sensible part: the air cooled from $t_1$ to $t_2$ at the entering humidity ratio $W_1$
[36] $Q_{latent} = \dot{m} (h(t_2, W_1) - h_2)$
$= 2.4 \times (48.026 - 32.985)$
$= 36.098$ kW
Latent part: the moisture removed from $W_1$ to $W_2$ at $t_2$
[37] $SHR = Q_{sensible} / (Q_{sensible} + Q_{latent})$
=
$= 0.52344$
Sensible heat ratio

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $T_{1,out}$ | 28 °C | Dry-bulb temperature, state 1 |
| $\varphi_1$ | 60 % | Relative humidity, state 1, as a quantity in % (for linking) |
| $T_{dp,1}$ | 19.51 °C | Dew point, state 1 |
| $T_{wb,1}$ | 22.07 °C | Wet-bulb temperature, state 1 |
| $W_1$ | 14.249 | Humidity ratio, state 1, in grams per kilogram of dry air |
| $h_1$ | 64.546 | Enthalpy, state 1, in kilojoules per kilogram of dry air |
| $T_{2,out}$ | 12 °C | Dry-bulb temperature, state 2 |
| $W_2$ | 8.2878 | Humidity ratio, state 2, in grams per kilogram of dry air |
| $h_2$ | 32.985 | Enthalpy, state 2, in kilojoules per kilogram of dry air |
| $p_{ws,1}$ | 3.782 kPa | Saturation pressure, state 1 |
| $p_{w,1}$ | 2.269 kPa | Vapour pressure, state 1 |
| $RH_1$ | 60 | Relative humidity, state 1, as a number from 0 to 100 |
| $v_1$ | 0.8727 m³/kg | Specific volume, state 1 |
| $\rho_1$ | 1.162 kg/m³ | Moist-air density, state 1 |
| $p_{ws,2}$ | 1.403 kPa | Saturation pressure, state 2 |
| $p_{w,2}$ | 1.332 kPa | Vapour pressure, state 2 |
| $RH_2$ | 95 | Relative humidity, state 2, as a number from 0 to 100 |
| $\varphi_2$ | 95 % | Relative humidity, state 2, as a quantity in % (for linking) |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $T_{dp,2}$ | 11.22 °C | Dew point, state 2 |
| $T_{wb,2}$ | 11.55 °C | Wet-bulb temperature, state 2 |
| $v_2$ | 0.8186 m³/kg | Specific volume, state 2 |
| $\rho_2$ | 1.232 kg/m³ | Moist-air density, state 2 |
| $m_w$ | 0.01431 kg/s | Condensate |
| $Q_{total}$ | 75.03 kW | Total cooling |
| $Q_{sensible}$ | 39.65 kW | Sensible cooling |
| $Q_{latent}$ | 36.10 kW | Latent cooling |
| SHR | 0.52344 | Sensible heat ratio |

## 6. Checks (governing first)

None.

## 7. Applicability

None.

Defaults used (inputs left empty):
$h_{steam}$: not entered, so the default of 2676 is used.

## 8. Verification

`Template bs.psychrometrics@1.4.0 Content hash 9ffd235faa1c3dae Engine calc-engine/1.0.0`
`Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0`

## 9. References

1. American Society of Heating, Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook—Fundamentals. ASHRAE.
2. Hyland RW, Wexler A. Formulations for the thermodynamic properties of the saturated phases of H2O from 173.15 K to 473.15 K. ASHRAE Trans. 1983;89(2A):500-519.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.